

Cloud Computing Lab

LAB # 01

LAB TITLE: *GitHub Identity, Student Portal Registration, and Ubuntu Server Installation*

Name	SANA NOOR BHATTI
Registration Number	FA24BI-SE-022
Section	B
Semester	Fifth (V)
Course Code	SE-665
GitHub Username	sananoorbhatti

Instructor: Engr. Waqas Saleem

Getting Started

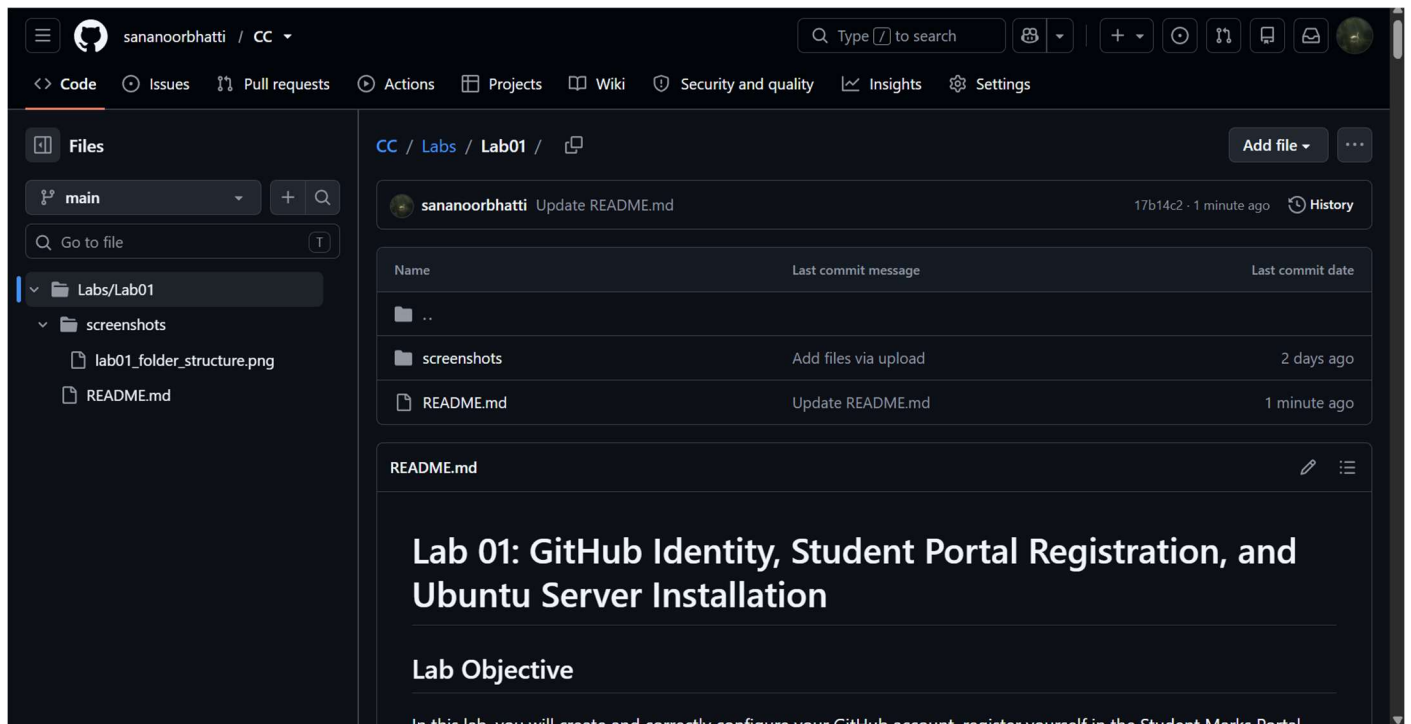


Figure 1 — lab01_folder_structure.png

Created the **Labs/Lab01/screenshots/** directory inside the **CC** repository before beginning the installation.

Task 1 — Create and Configure Your GitHub Account

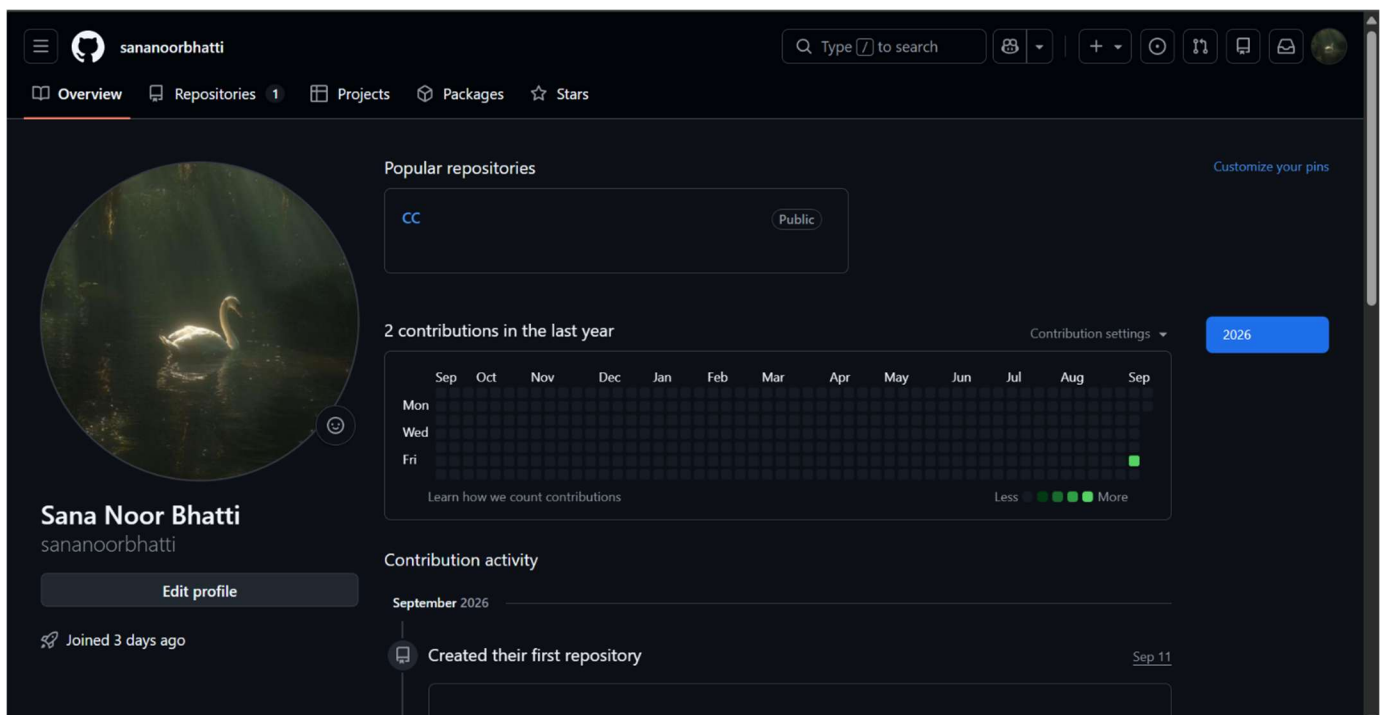


Figure 2 — github_profile.png

A personal GitHub account was created showing the complete name (**Sana Noor Bhatti**) and professional GitHub username (**sananoorbhatti**).

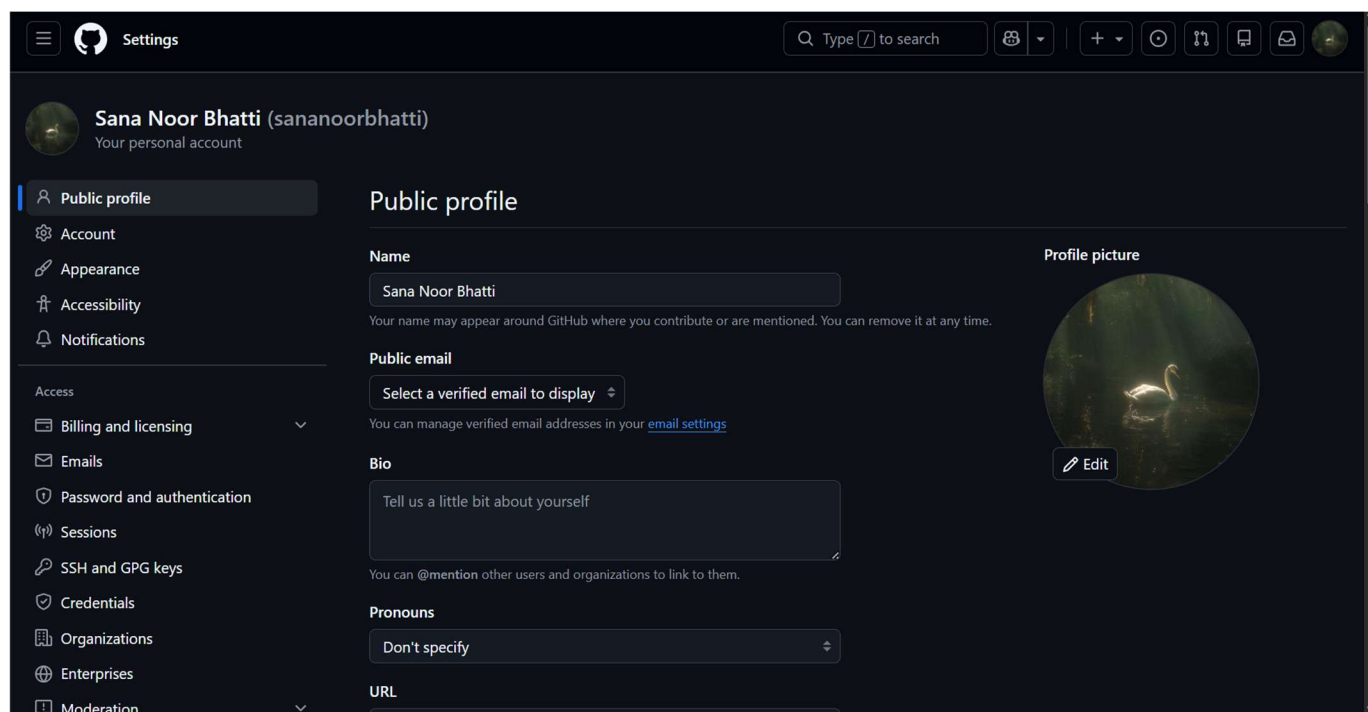


Figure 3 — github_actual_name.png

The GitHub Public Profile settings showing the complete actual name entered in the Name field.

Task 2 — Register in the Student Marks Portal

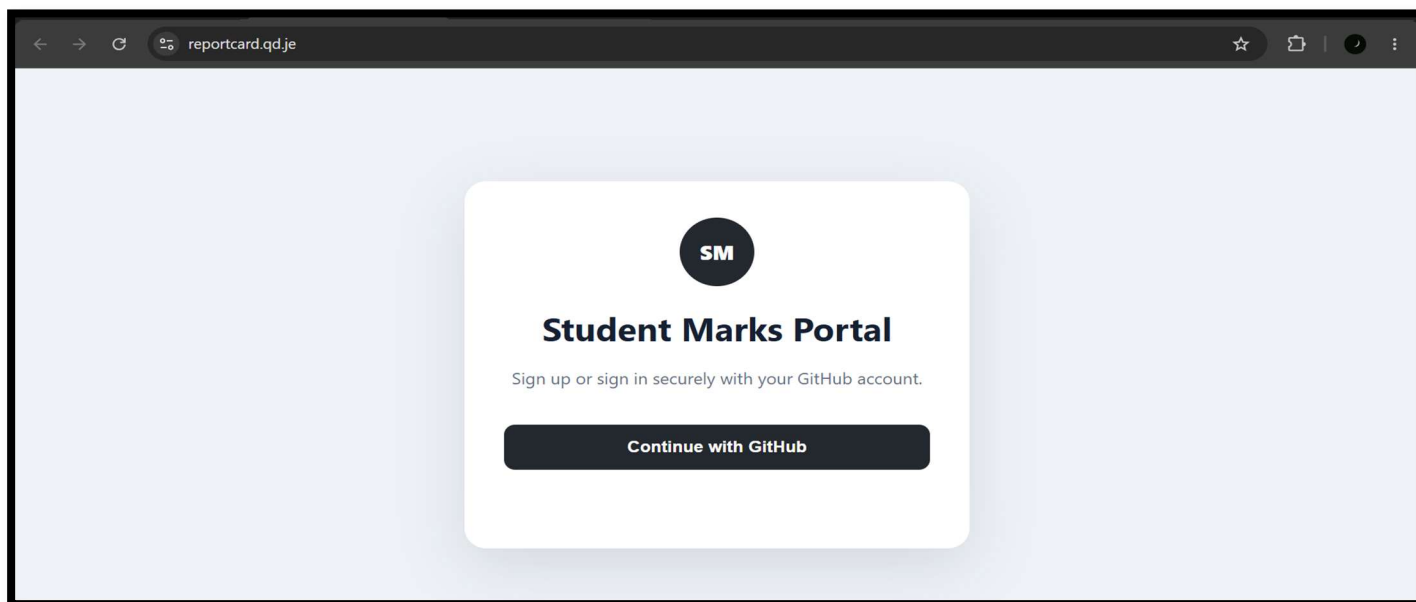


Figure 4 — portal_login_page.png

The Student Marks Portal login page is displayed, providing the option to continue using the GitHub

The screenshot shows a web browser window with the URL `reportcard.qd.je`. The page has a light blue background. In the center, there is a white card with a dark blue circle containing the text "ID". Below this, the heading "Complete registration" is displayed in bold. Underneath, a message states: "Enter your registration number, course and section. The enrollment requires approval." The form contains three input fields: "Registration number" with the value "2024-BSE-22", "Course" with the value "SE-665 — Cloud Computing", and "Section" with the value "Section B". Below these fields is a dark blue button labeled "Submit enrollment" and a light blue button labeled "Sign out". A small text note below the registration number field says "Allowed: 2024-BSE-01 to 2024-BSE-99."

Figure 5 — *portal_github_profile.png*

The Student Marks Portal displaying the GitHub account information after successful GitHub authentication.

The screenshot shows the "STUDENT MARKS PORTAL" with the heading "Course enrollments". In the top right corner, there is a "Sign out" button. Below the heading, there is a user profile card for "Sana Noor Bhatti" with a red and white pixelated avatar and the registration number "2024-BSE-22". To the right of the profile card is a dark blue button labeled "+ Enroll in another course". Below the profile card, there is a section titled "ENROLLMENT STATUS" with the heading "Your courses". Under this heading, there is a card for "Cloud Computing (SE-665) Section B" with the status "Waiting for administrator approval" and a yellow button labeled "Pending approval".

Figure 6 — *portal_enrollment_submitted.png*

The completed enrollment information is displayed, with the enrollment status shown as “**Pending Approval.**”

Task 3 — Download VMware Workstation Pro and Ubuntu Server

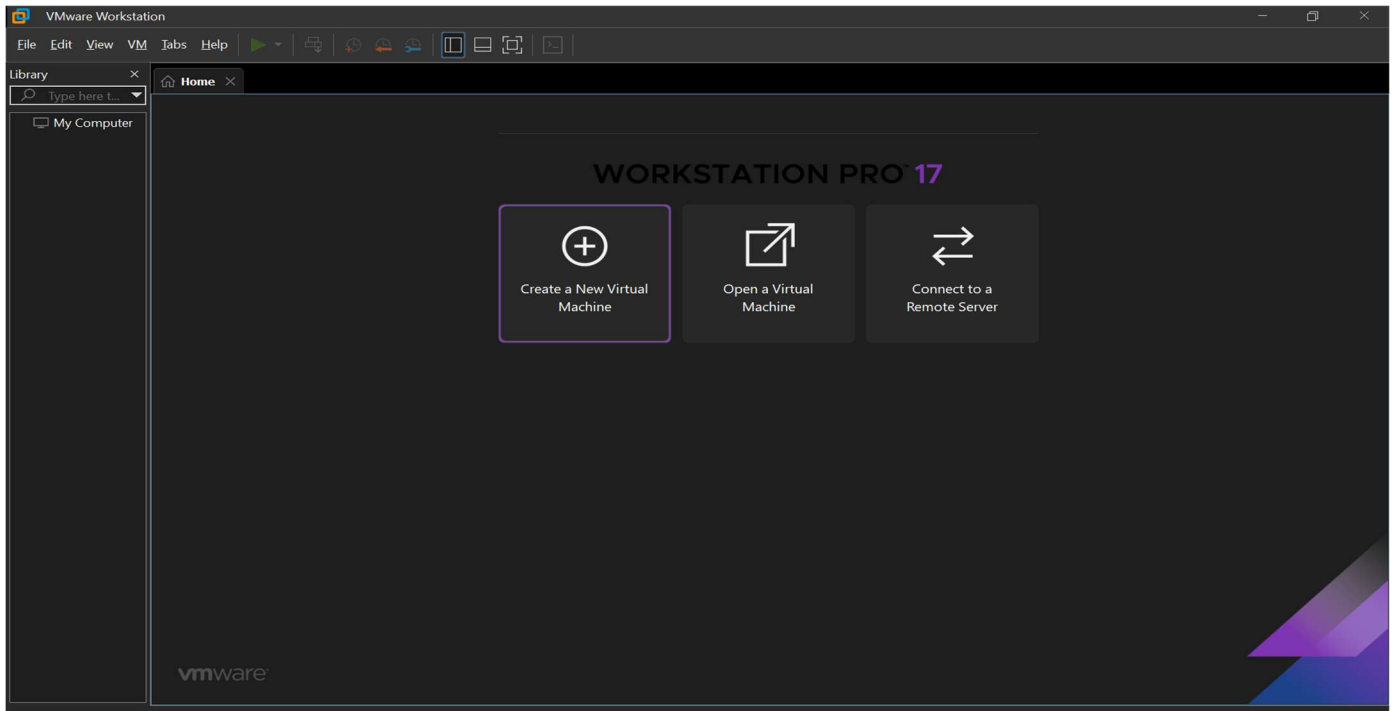


Figure 7 — vmware_workstation.png

VMware Workstation Pro is installed and running successfully on the Windows computer.

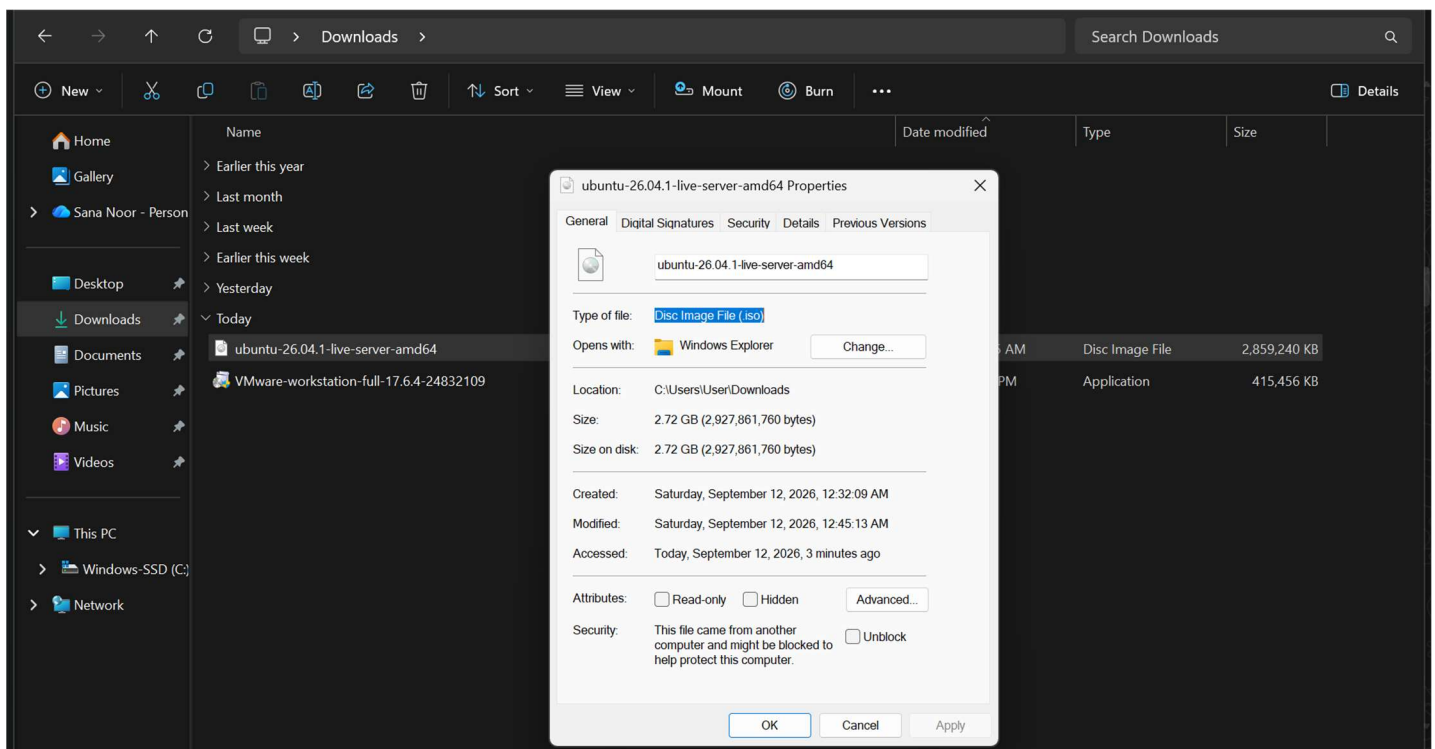


Figure 8 — ubuntu_server_iso.png

The downloaded **Ubuntu Server LTS ISO** file is shown, providing the installation media required for the virtual machine.

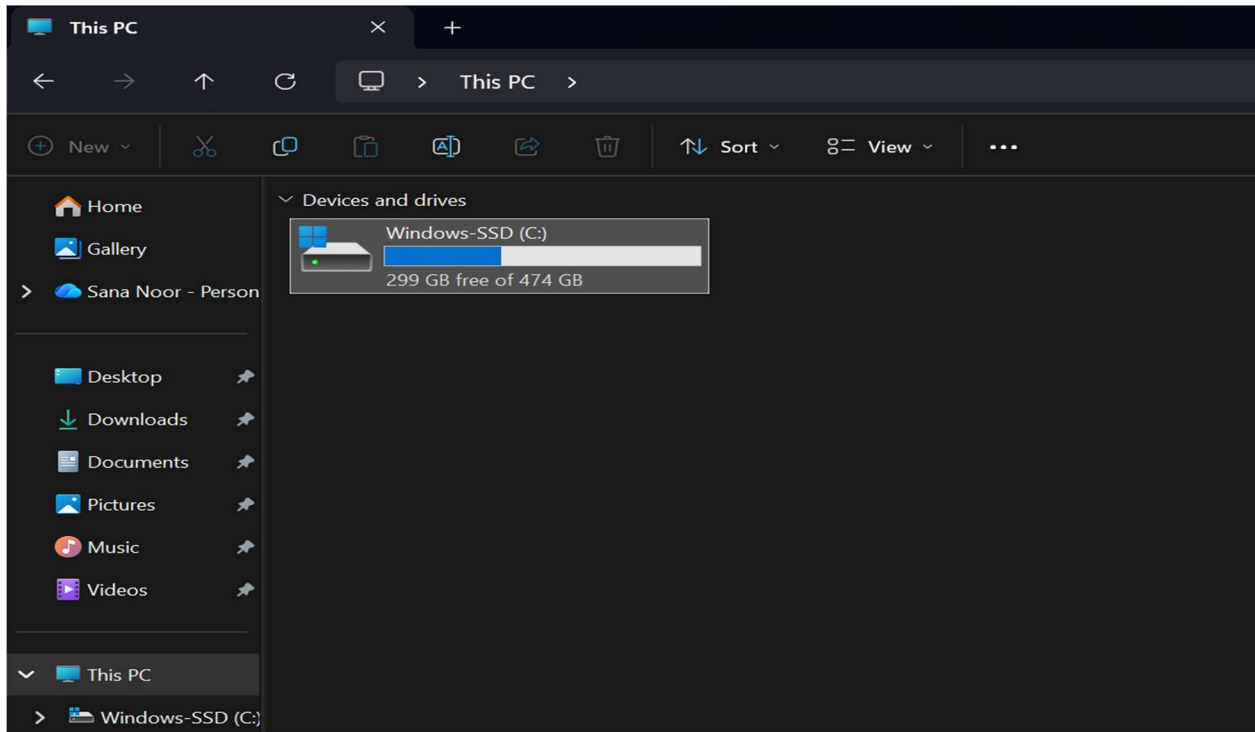


Figure 9 — available_storage.png

The available Windows storage is shown and confirms that sufficient free space is available for the Ubuntu Server virtual machine.

Task 4 — Create the Ubuntu Server Virtual Machine

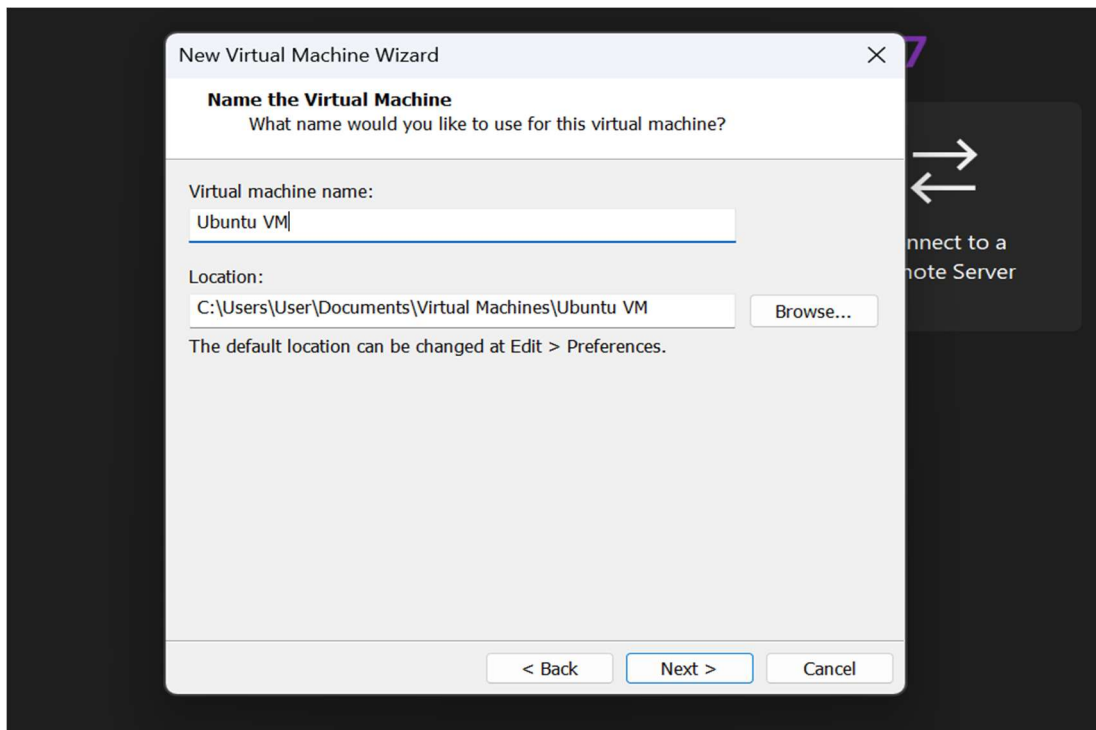


Figure 10 — vm_creation_wizard.png

The VMware New Virtual Machine Wizard is opened to begin creating the Ubuntu Server virtual machine.



Figure 11 — *vm_typical_configuration.png*

The **Typical (recommended)** virtual machine configuration is selected in VMware Workstation.

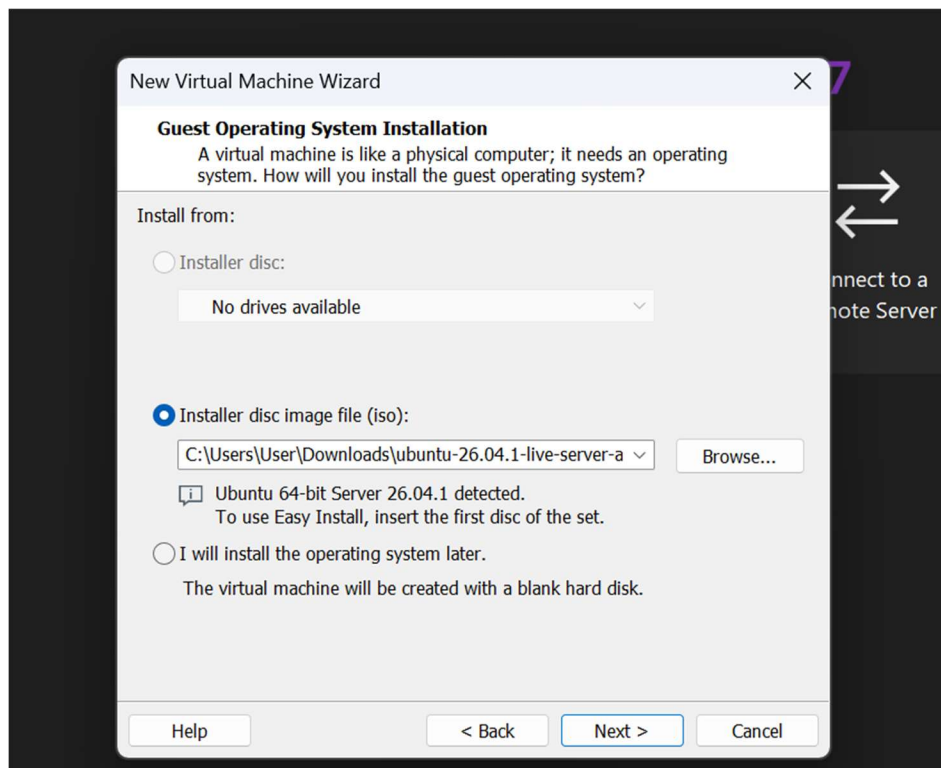


Figure 12 — *vm_ubuntu_iso_selected.png*

The Ubuntu Server ISO is selected as the installation disc image for the virtual machine.

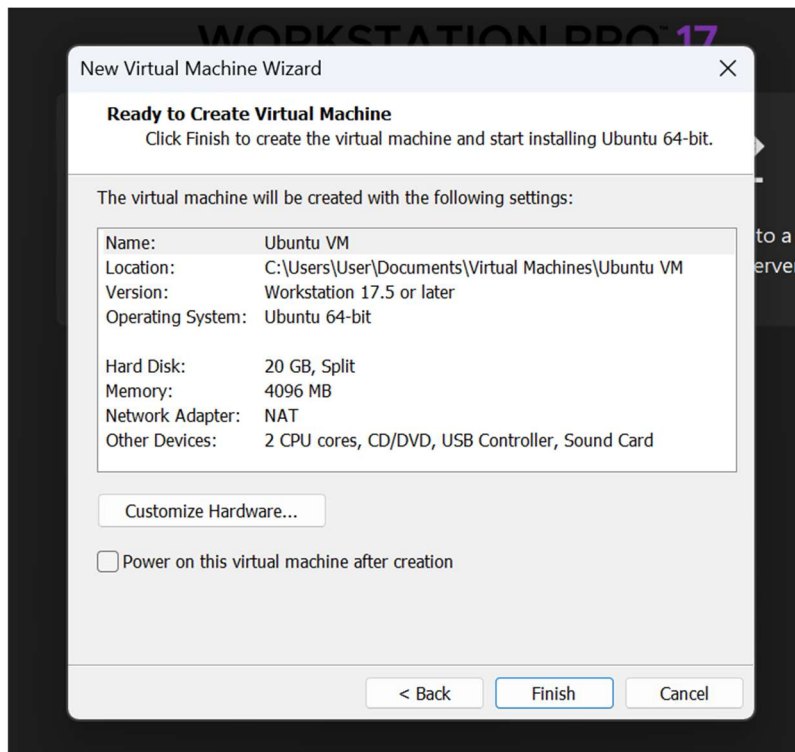


Figure 13 — *vm_configuration_summary.png*

The final virtual machine configuration is reviewed before completing the VMware virtual machine creation process.

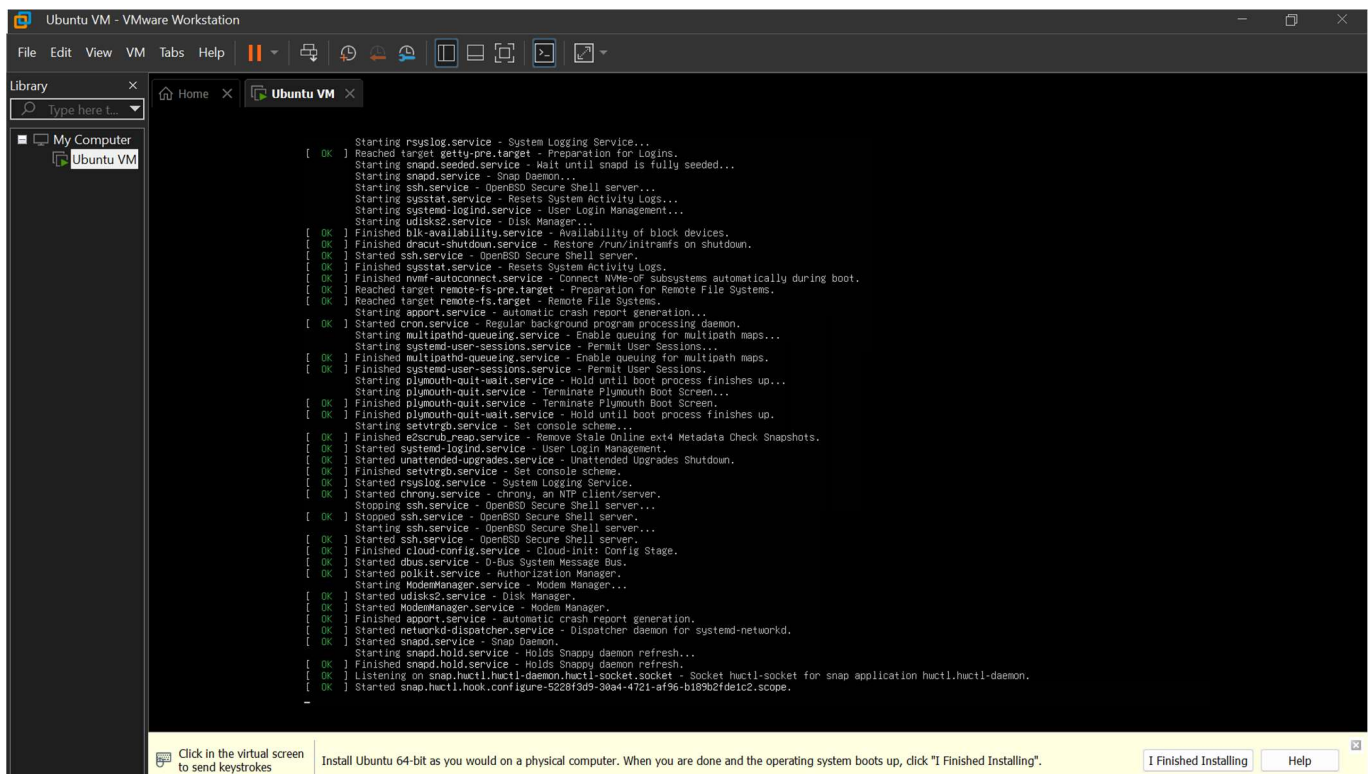


Figure 14 — *ubuntu_installer_boot.png*

The Ubuntu Server installer **boot screen** is displayed after powering on the newly created virtual machine.

Task 5 — Choose the Installation Language

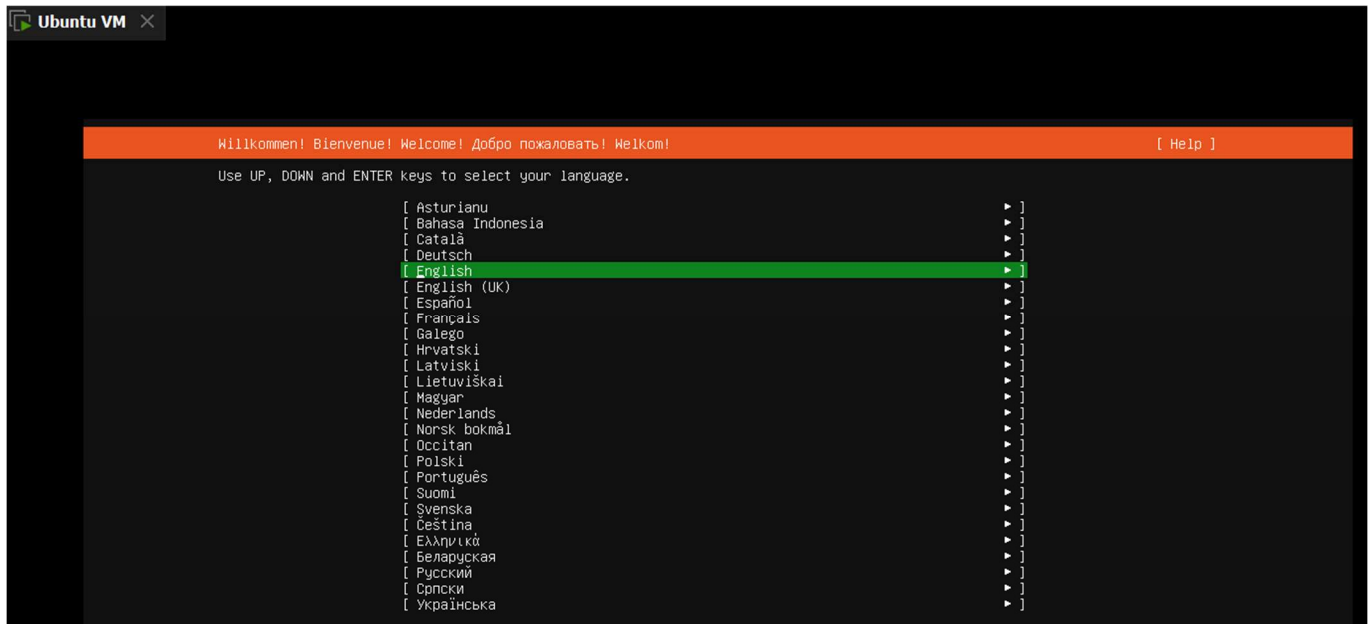


Figure 15 — *ubuntu_language.png*

English was selected as the installation language in Ubuntu Server installer.

Task 6 — Configure the Keyboard Layout

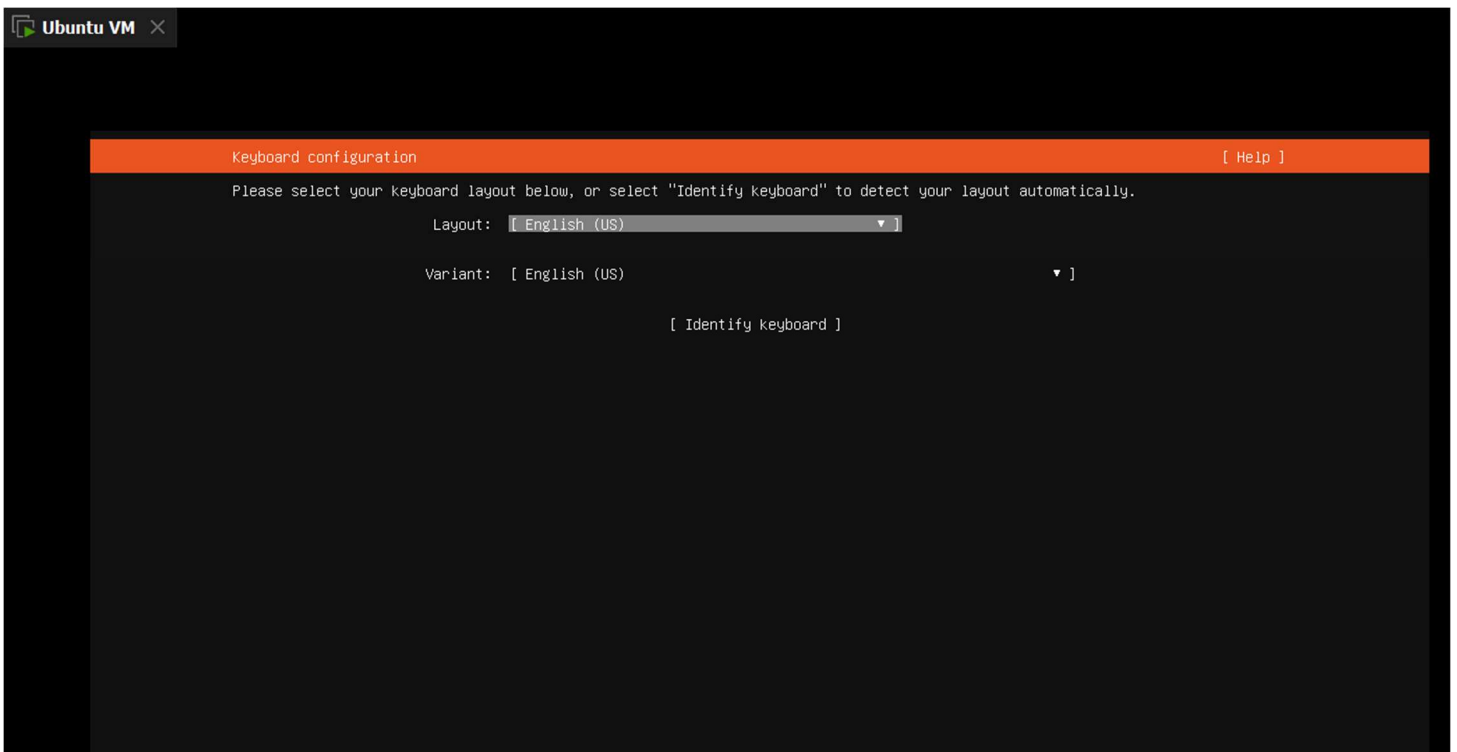


Figure 16 — *ubuntu_keyboard_layout.png*

English (US) keyboard layout matching the computer keyboard is selected in the Ubuntu Server installer.

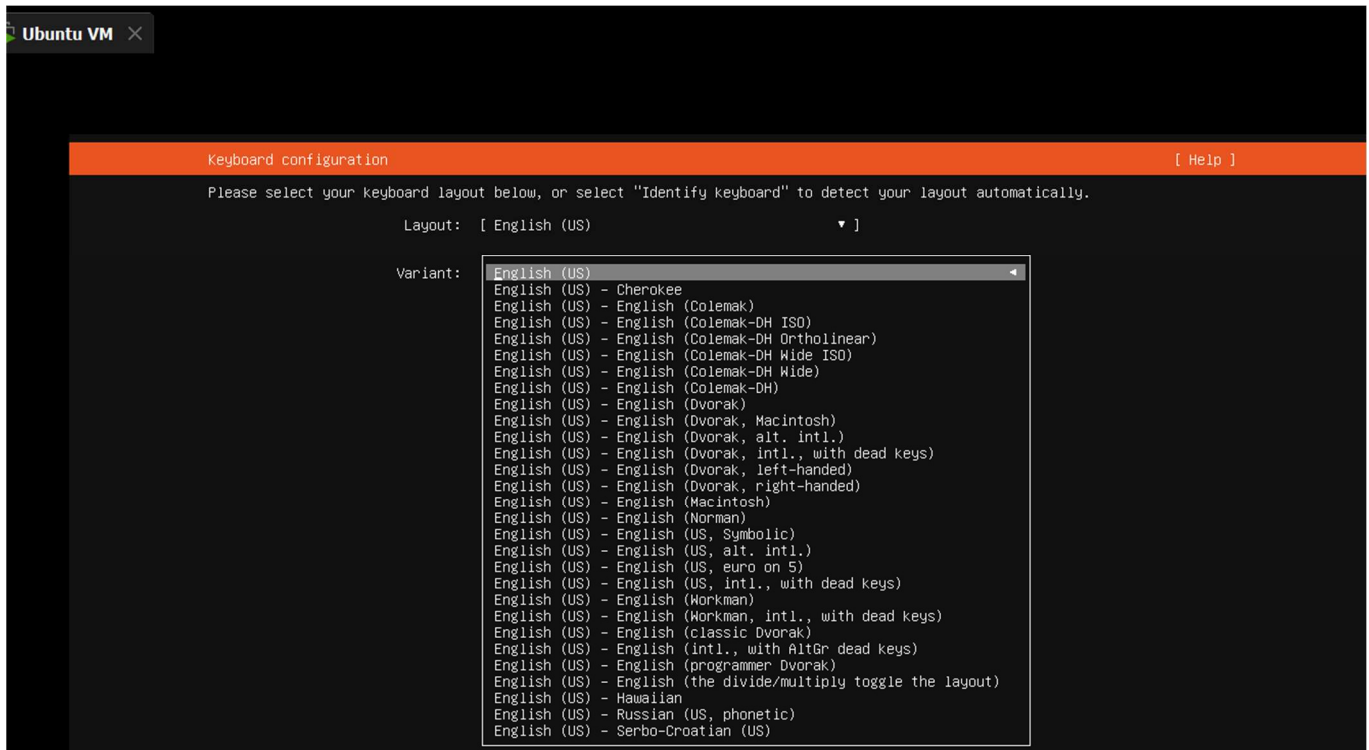


Figure 17 — *ubuntu_keyboard_variant.png*

English (US) keyboard variant is selected to complete the keyboard configuration.

Task 7 — Choose the Ubuntu Server Installation Type

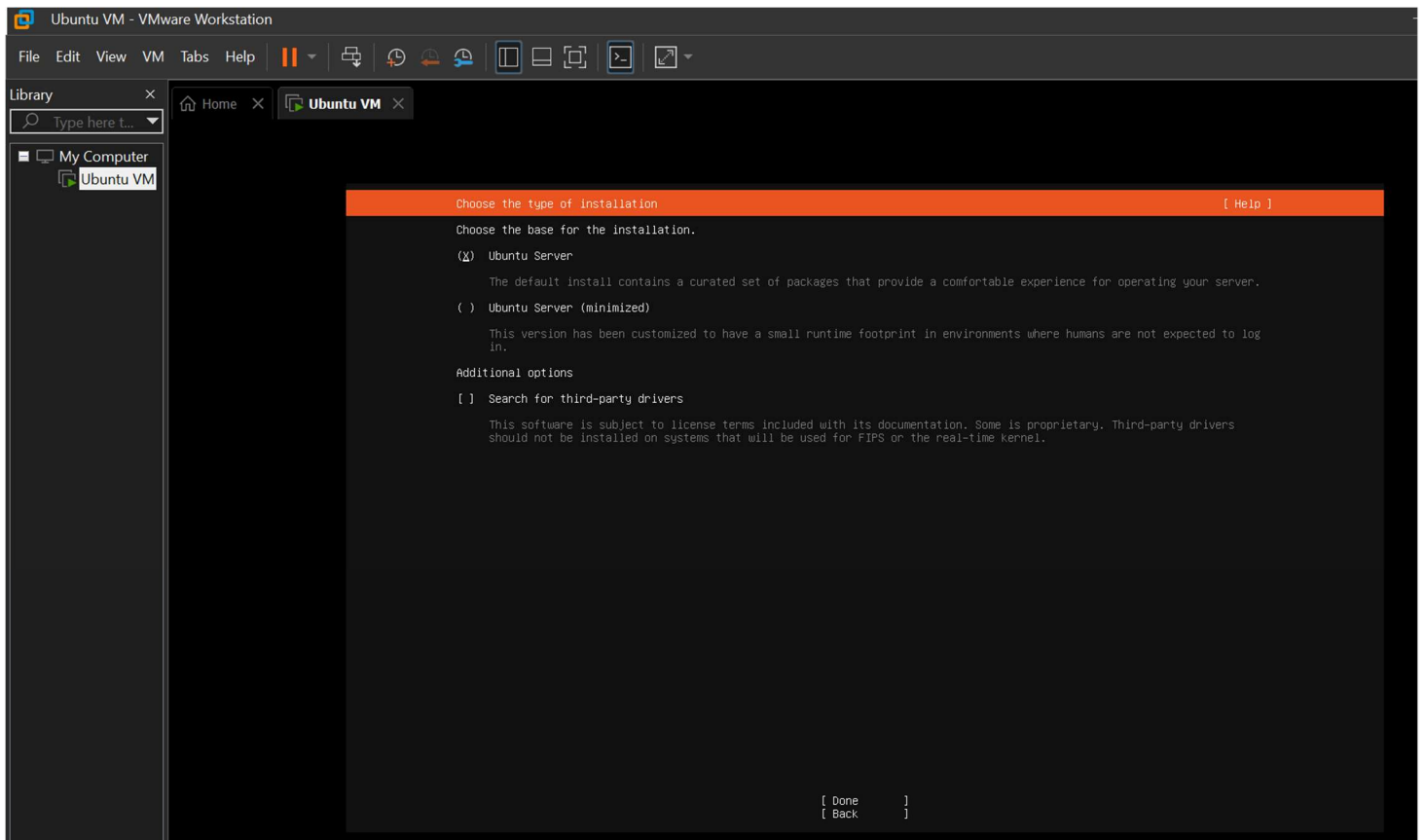


Figure 18 — *ubuntu_installation_type.png*

The Ubuntu Server (normal installation) option was selected, rather than the minimized or MAAS variants.

Task 8 — Configure the Network

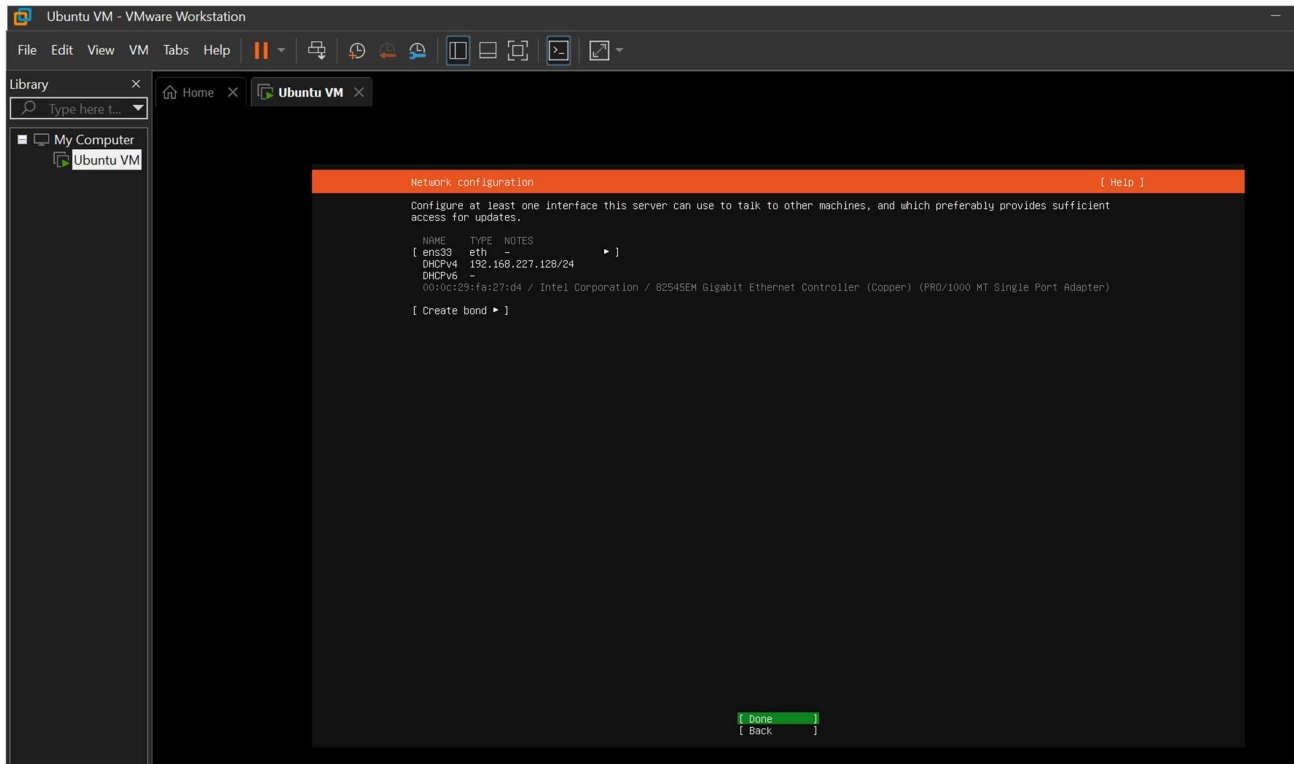


Figure 19 — *ubuntu_network_interface.png*

The network interface detected (ens33) by the Ubuntu Server installer is displayed and reviewed.

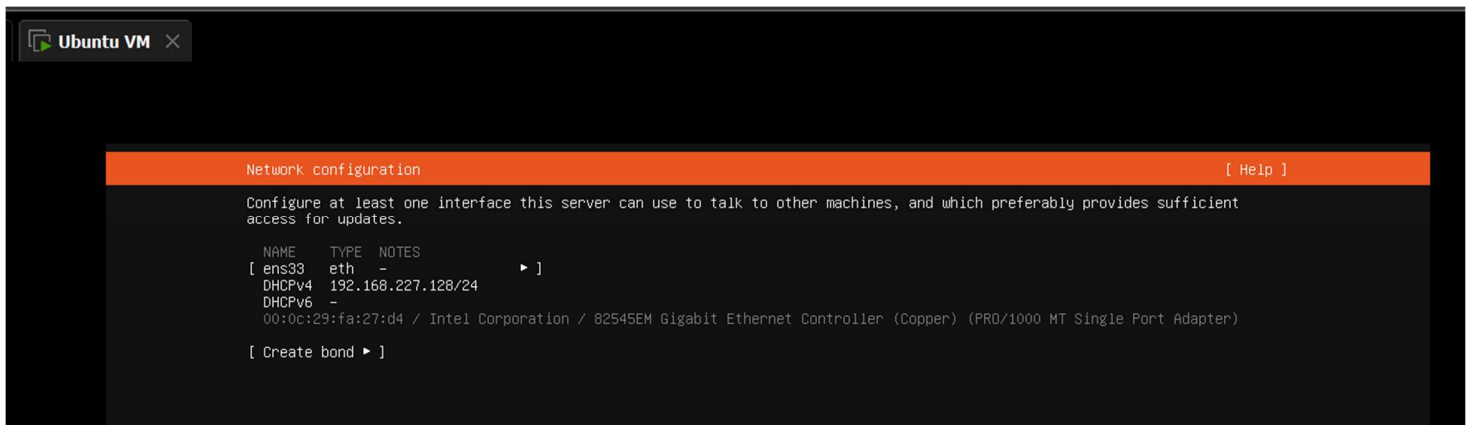


Figure 20 — *ubuntu_installer_network.png*

The network interface has successfully received an IP address automatically through **DHCPv4** which is **192.168.227.128**.

Task 9 — Configure Storage

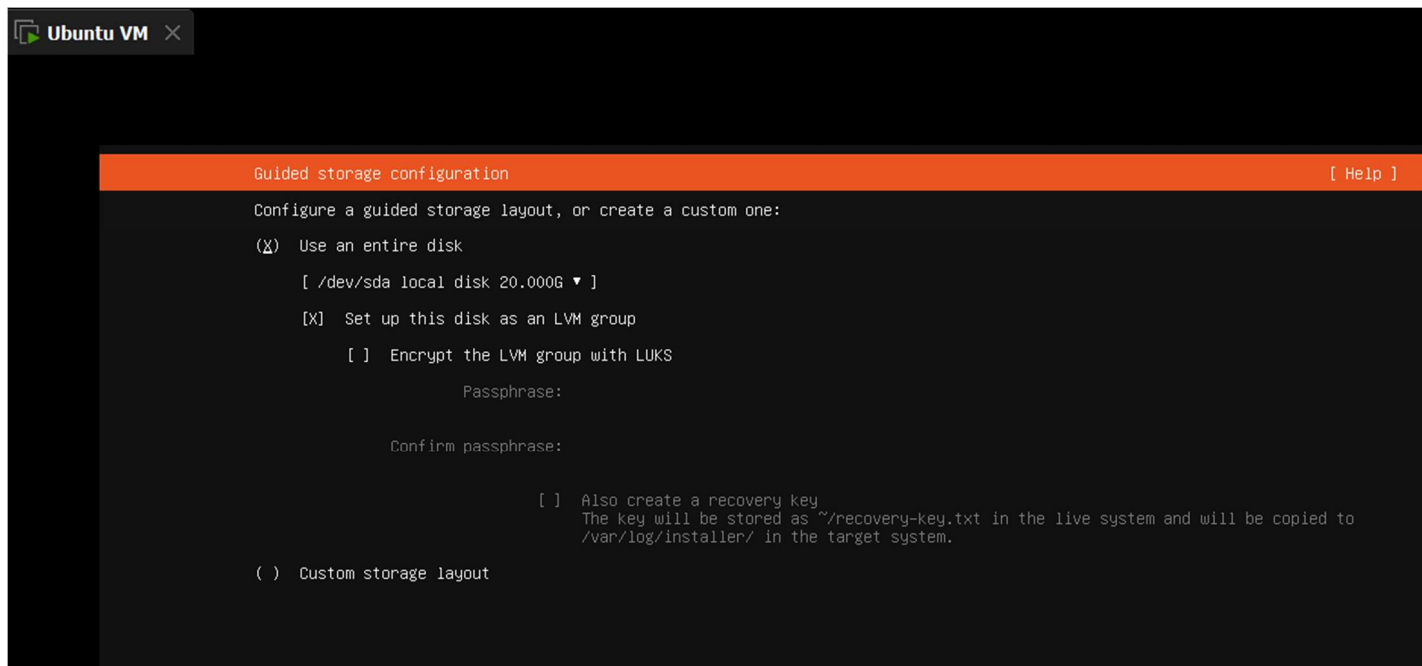


Figure 21 — ubuntu_guided_storage.png

The **guided storage option** was used so the installer would use the entire VMware virtual disk

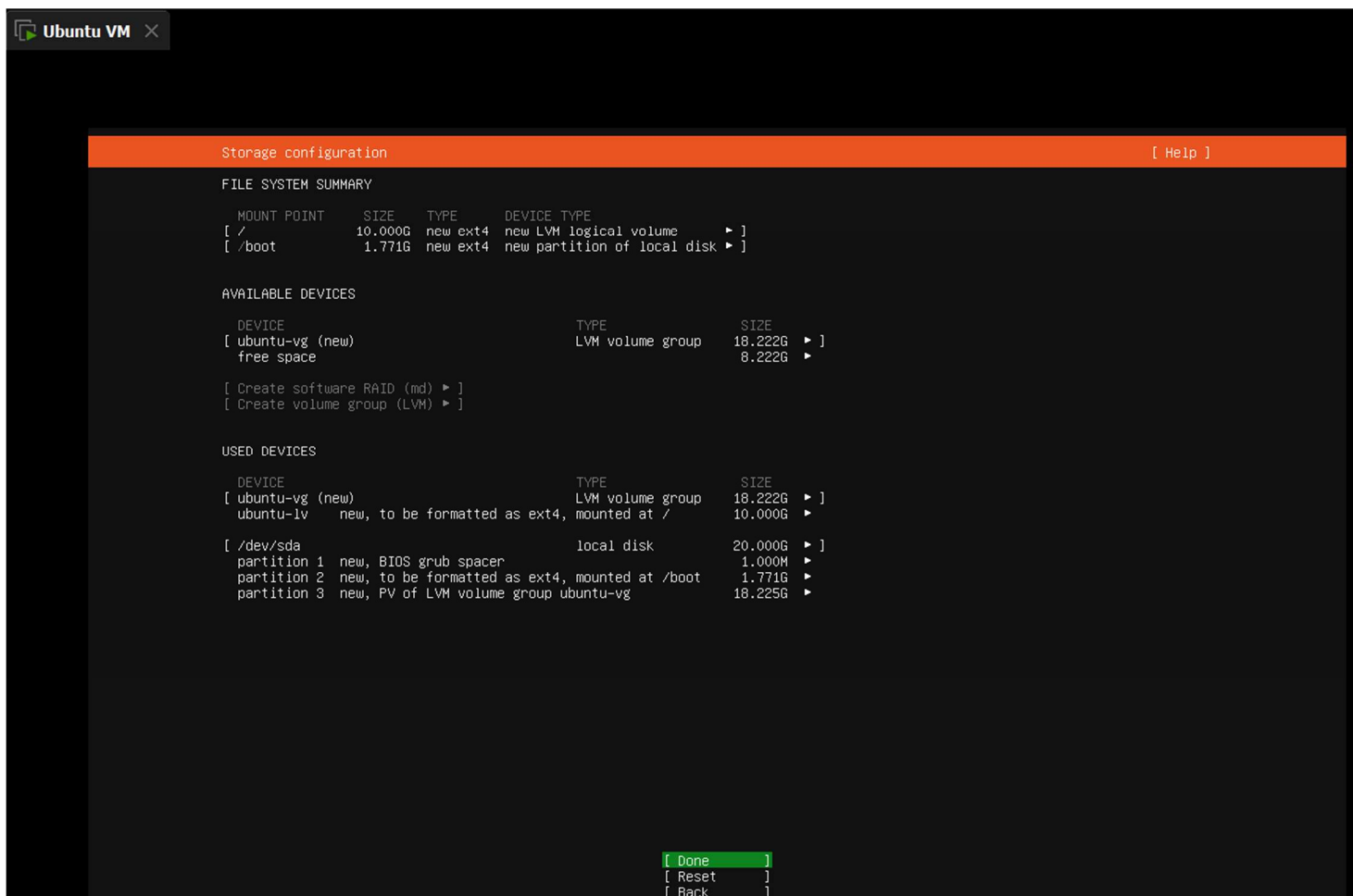


Figure 22 — ubuntu_storage_configuration.png

The proposed storage configuration is reviewed before continuing with the Ubuntu Server installation.

Task 10 — Select the Installation Disk

```
DEVICE                                TYPE                                SIZE
[ ubuntu-vg (new)                     LVM volume group                  18.222G ▶ ]
  ubuntu-lv      new, to be formatted as ext4, mounted at /    10.000G ▶ ]

[ /dev/sda                             local disk                        20.000G ▶ ]
  partition 1   new, BIOS grub spacer                            1.000M ▶
  partition 2   new, to be formatted as ext4, mounted at /boot  1.771G ▶
  partition 3   new, PV of LVM volume group ubuntu-vg           18.225G ▶
```

Figure 23 — ubuntu_selected_disk.png

The **VMware virtual disk (/dev/sda)** created for the Ubuntu Server virtual machine is selected as the installation disk.

USED DEVICES

```
DEVICE                                TYPE                                SIZE
[ ubuntu-vg (new)                     LVM volume group                  18.222G ▶ ]
  ubuntu-lv      new, to be formatted as ext4, mounted at /    10.000G ▶ ]

[ /dev/sda                             local disk                        20.000G ▶ ]
  partition 1   new, BIOS grub spacer                            1.000M ▶
  partition 2   new, to be formatted as ext4, mounted at /boot  1.771G ▶
  partition 3   new, PV of LVM volume group ubuntu-vg           18.225G ▶
```

Figure 24 — ubuntu_disk_details.png

The selected virtual disk **name** and displayed disk **size(20.000G)** are reviewed to verify the correct VMware disk.

Task 11 — Review the Partitions

```
[ /dev/sda                             local disk                        20.000G ▶ ]
  partition 1   new, BIOS grub spacer                            1.000M ▶
  partition 2   new, to be formatted as ext4, mounted at /boot  1.771G ▶
  partition 3   new, PV of LVM volume group ubuntu-vg           18.225G ▶
```

Figure 25 — ubuntu_partition_layout.png

The **partition layout** proposed by the Ubuntu Server installer is reviewed before applying the storage configuration.

Task 12 — Confirm the Storage Changes

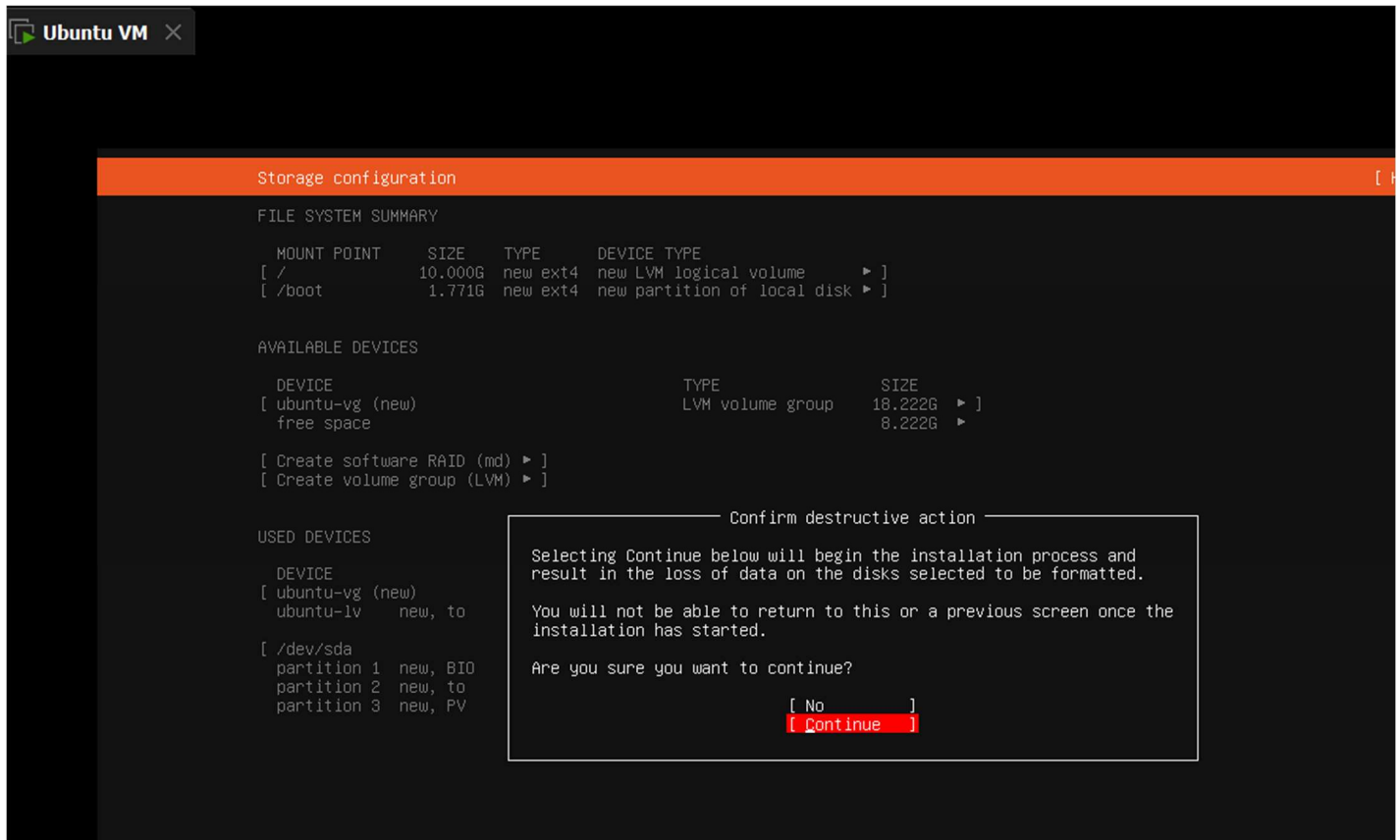


Figure 26 — *ubuntu_storage_warning.png*

The final destructive-action warning is displayed before applying the proposed storage changes to the virtual disk.

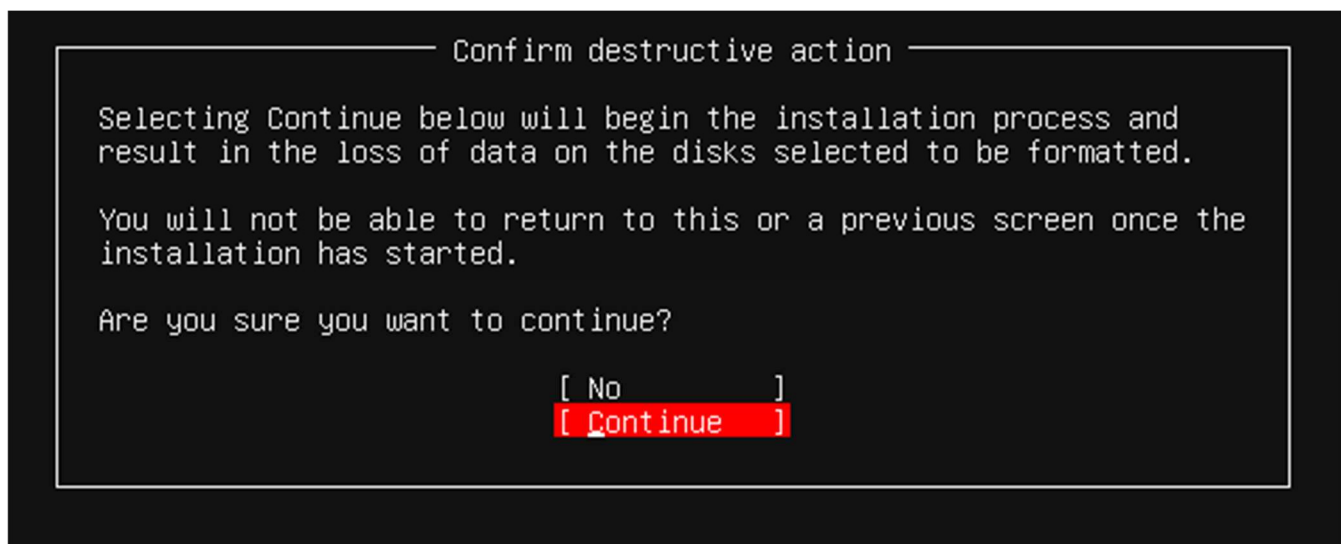
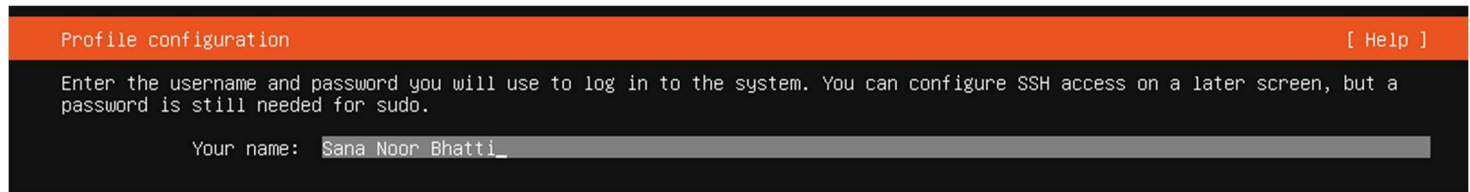


Figure 27 — *ubuntu_storage_confirmed.png*

The storage configuration is confirmed after verifying that the correct VMware virtual disk is selected.

Task 13 — Configure the Ubuntu Profile



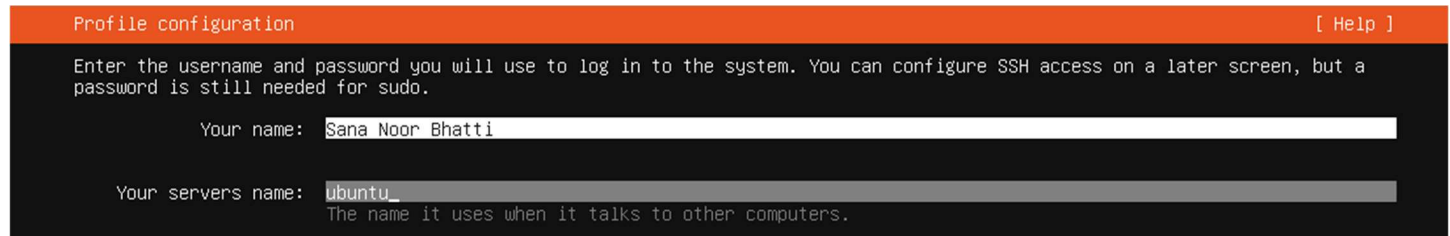
Profile configuration [Help]

Enter the username and password you will use to log in to the system. You can configure SSH access on a later screen, but a password is still needed for sudo.

Your name:

Figure 28 — *ubuntu_actual_name.png*

The complete actual name is entered in the Your name field of the Ubuntu profile configuration.



Profile configuration [Help]

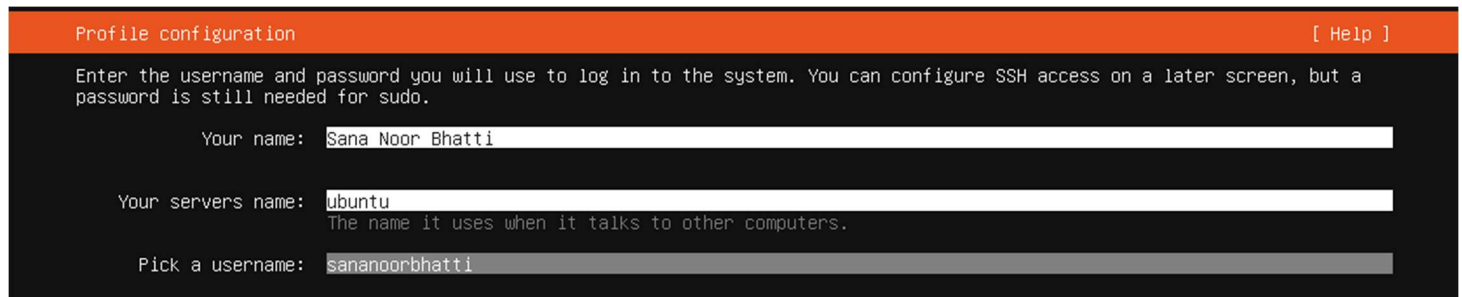
Enter the username and password you will use to log in to the system. You can configure SSH access on a later screen, but a password is still needed for sudo.

Your name:

Your servers name:
The name it uses when it talks to other computers.

Figure 29 — *ubuntu_server_name.png*

The Ubuntu server name is set to “ubuntu” according to the required instructions.



Profile configuration [Help]

Enter the username and password you will use to log in to the system. You can configure SSH access on a later screen, but a password is still needed for sudo.

Your name:

Your servers name:
The name it uses when it talks to other computers.

Pick a username:

Figure 30 — *ubuntu_username.png*

The Ubuntu username is configured to exactly match the GitHub username.



Profile configuration [Help]

Enter the username and password you will use to log in to the system. You can configure SSH access on a later screen, but a password is still needed for sudo.

Your name:

Your servers name:
The name it uses when it talks to other computers.

Pick a username:

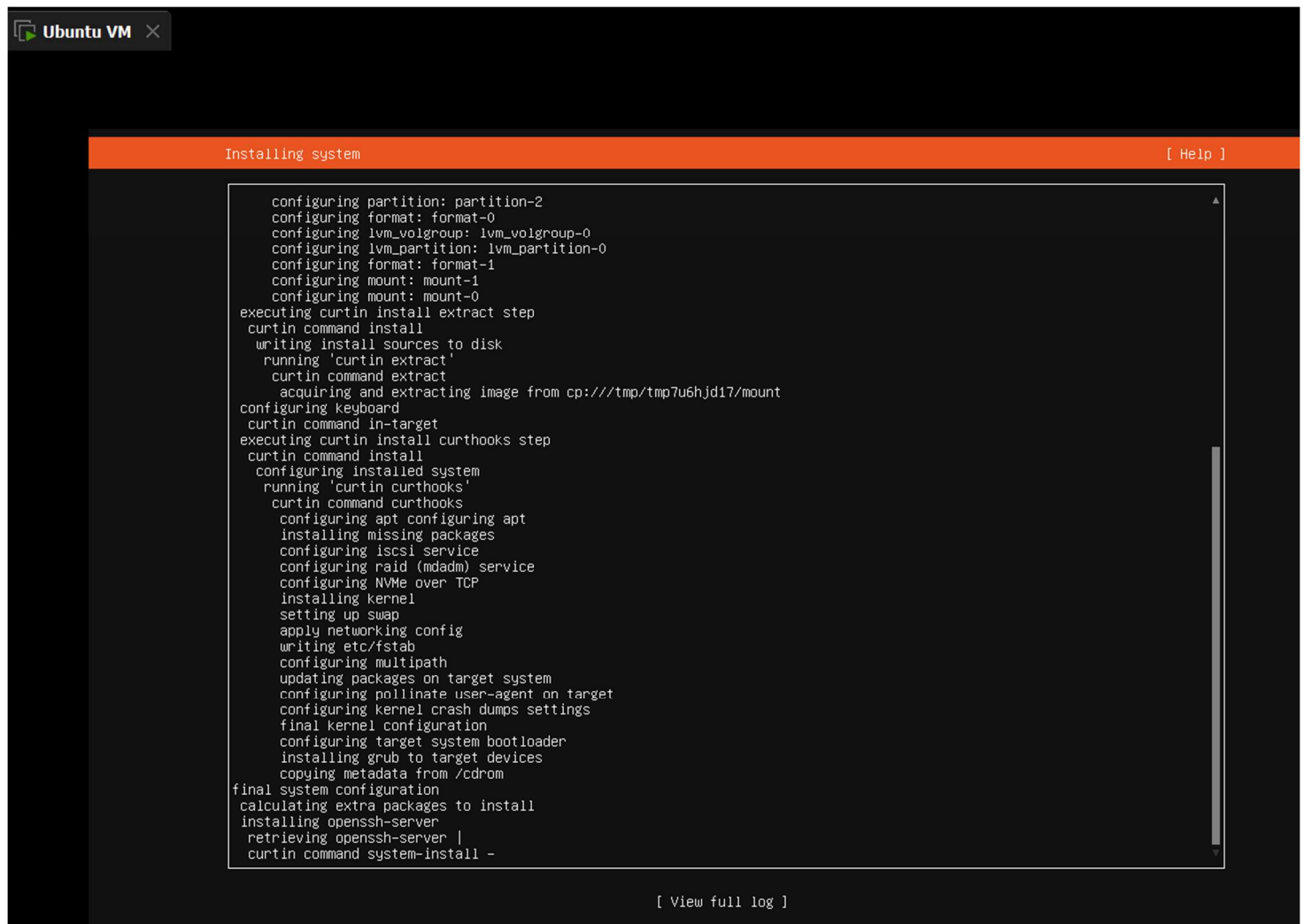
Choose a password:

Confirm your password:

Figure 31 — *ubuntu_profile_setup.png*

The completed Ubuntu profile configuration is reviewed with the required identity information entered.

Task 14 — Complete the Software Installation



The screenshot shows the Ubuntu VM installer interface. At the top, there is a title bar with the Ubuntu logo and the text 'Ubuntu VM'. Below this is a progress bar with the text 'Installing system' and a '[Help]' link on the right. The main area of the window displays a list of configuration steps in a terminal-like font. The steps include configuring partitions, formats, LVM volumes, mounts, and executing various curtin commands for installation, extraction, and system configuration. The list ends with 'final system configuration' and 'calculating extra packages to install'. A '[View full log]' link is located at the bottom right of the terminal area.

```
configuring partition: partition-2
configuring format: format-0
configuring lvm_volgroup: lvm_volgroup-0
configuring lvm_partition: lvm_partition-0
configuring format: format-1
configuring mount: mount-1
configuring mount: mount-0
executing curtin install extract step
curtin command install
writing install sources to disk
running 'curtin extract'
curtin command extract
acquiring and extracting image from cp:///tmp/tmp7u6hjd17/mount
configuring keyboard
curtin command in-target
executing curtin install curthooks step
curtin command install
configuring installed system
running 'curtin curthooks'
curtin command curthooks
configuring apt configuring apt
installing missing packages
configuring iscsi service
configuring raid (mdadm) service
configuring NVMe over TCP
installing kernel
setting up swap
apply networking config
writing etc/fstab
configuring multipath
updating packages on target system
configuring pollinate user-agent on target
configuring kernel crash dumps settings
final kernel configuration
configuring target system bootloader
installing grub to target devices
copying metadata from /cdrom
final system configuration
calculating extra packages to install
installing openssh-server
retrieving openssh-server |
curtin command system-install -
```

[View full log]

Figure 32 — *ubuntu_installation_progress.png*

The Ubuntu Server installer is installing the required software and system components.

Task 15 — Reboot and Verify the Ubuntu Identity

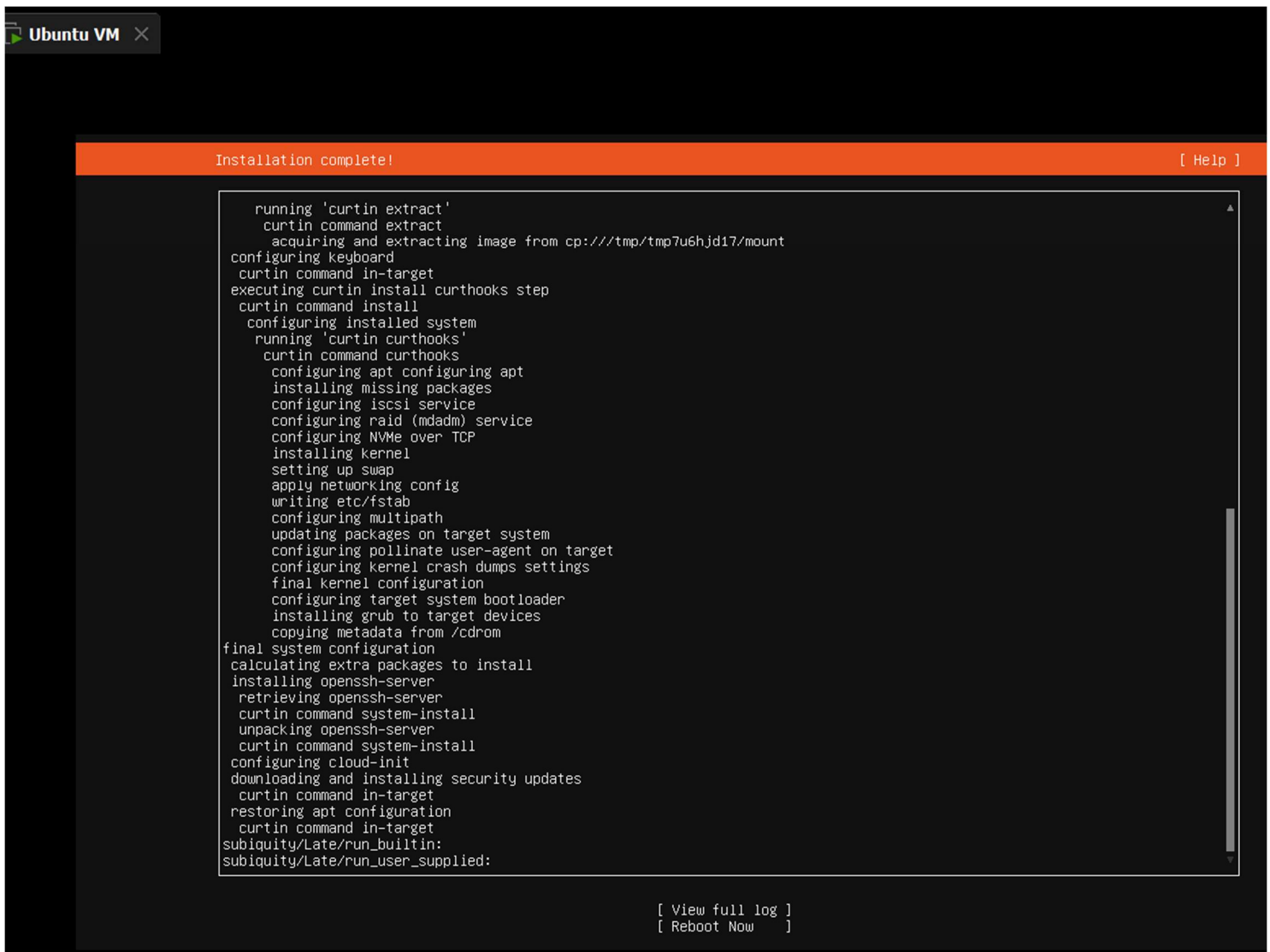


Figure 33 — ubuntu_installation_complete.png

The Ubuntu Server installation has completed successfully and the system is ready to reboot.

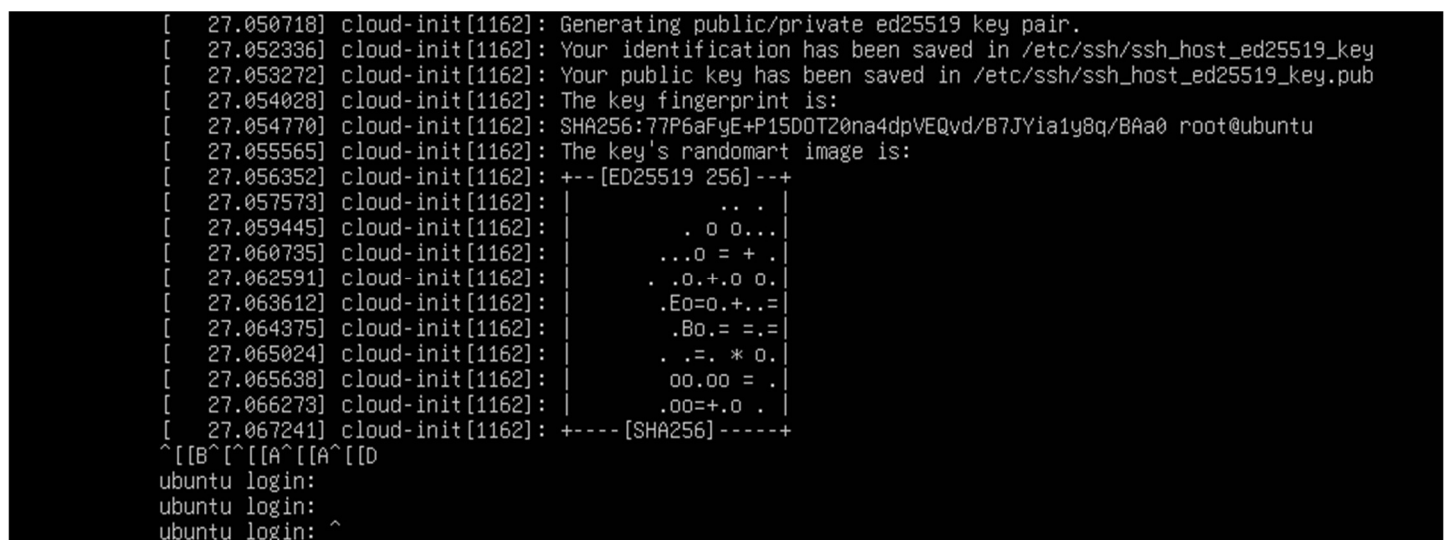


Figure 34 — ubuntu_login_screen.png

The installed Ubuntu Server displays the login screen after the system starts after the first boot.

```
ubuntu login:
ubuntu login: sananoorbhatti
Password:
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-31-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Sat Sep 12 07:51:50 PM UTC 2026

System load: 0.05          Processes:            215
Usage of /:  33.8% of 9.75GB Users logged in:          0
Memory usage: 11%         IPv4 address for ens33: 192.168.227.129
Swap usage:  0%

Expanded Security Maintenance for Applications is not enabled.

6 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

sananoorbhatti@ubuntu:~$
```

Figure 35 — *ubuntu_terminal_login.png*

Successfully logs in to the Ubuntu Server and accessing the terminal.

```
Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

sananoorbhatti@ubuntu:~$
```

Figure 36 — *ubuntu_identity_verified.png*

The Ubuntu terminal prompt confirming the **sananoorbhatti@ubuntu:~\$** format (the username matches the GitHub username and that the server name is **ubuntu**.)

Task 16 — Find the Ubuntu Server IP Address

```

sananoorbhatti@ubuntu:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 00:0c:29:fa:27:d4 brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    altname enx000c29fa27d4
    inet 192.168.227.129/24 metric 100 brd 192.168.227.255 scope global dynamic ens33
        valid_lft 1263sec preferred_lft 1263sec
    inet6 fe80::20c:29ff:fefa:27d4/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
sananoorbhatti@ubuntu:~$ _

```

Figure 37 — ubuntu_ip_addr_command.png

The "ip addr" command is executed and its output is displayed.

```

sananoorbhatti@ubuntu:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 00:0c:29:fa:27:d4 brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    altname enx000c29fa27d4
    inet 192.168.227.129/24 metric 100 brd 192.168.227.255 scope global dynamic ens33
        valid_lft 1263sec preferred_lft 1263sec
    inet6 fe80::20c:29ff:fefa:27d4/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
sananoorbhatti@ubuntu:~$

```

Figure 38 — ubuntu_ip_address.png

The active network interface shows the Ubuntu Server's correct IPv4 address (192.168.227.129), while the loopback address 127.0.0.1 is excluded.

Task 17 — Connect to Ubuntu Server from Windows

```

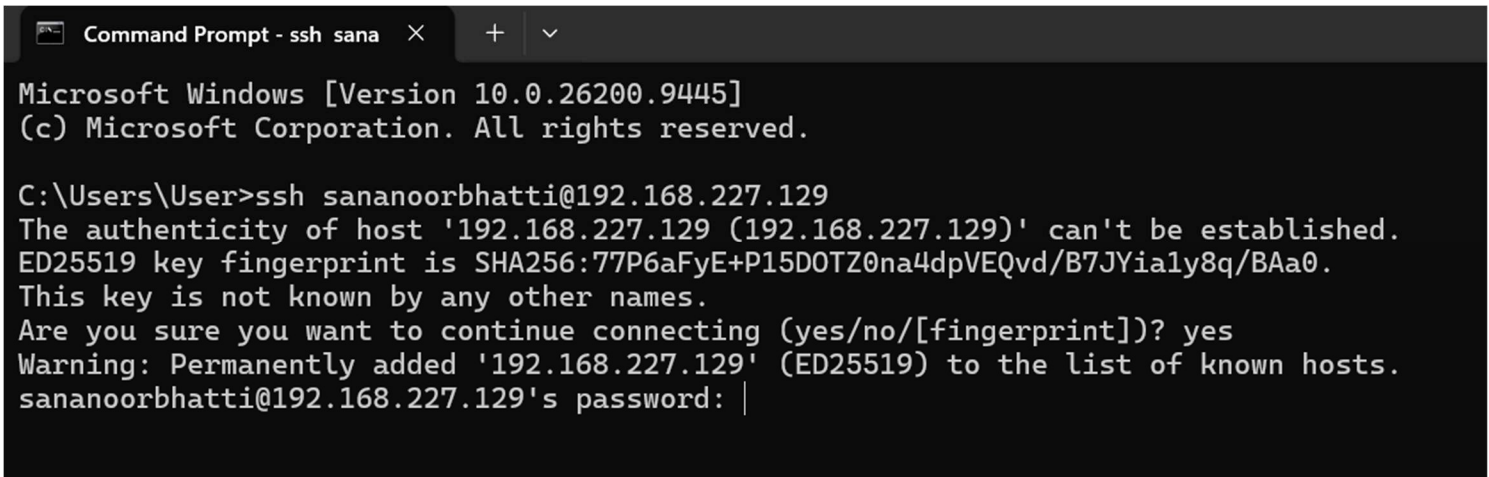
Microsoft Windows [Version 10.0.26200.9445]
(c) Microsoft Corporation. All rights reserved.

C:\Users\User>ssh sananoorbhatti@192.168.227.129
The authenticity of host '192.168.227.129 (192.168.227.129)' can't be established.
ED25519 key fingerprint is SHA256:77P6aFyE+P15DOTZ0na4dpVEQvd/B7JYia1y8q/BAa0.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? |

```

Figure 39 — windows_ssh_command.png

The **SSH command** is executed from Windows Command Prompt using the Ubuntu username and server IP address to initiate the remote connection.



```
Command Prompt - ssh sana X + v
Microsoft Windows [Version 10.0.26200.9445]
(c) Microsoft Corporation. All rights reserved.

C:\Users\User>ssh sananoorbhatti@192.168.227.129
The authenticity of host '192.168.227.129 (192.168.227.129)' can't be established.
ED25519 key fingerprint is SHA256:77P6aFyE+P15DOTZ0na4dpVEQvd/B7JYia1y8q/BAa0.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.227.129' (ED25519) to the list of known hosts.
sananoorbhatti@192.168.227.129's password: |
```

Figure 40 — windows_ssh_fingerprint.png

The **SSH server fingerprint is accepted**, allowing the Windows system to establish trust with the Ubuntu Server.

```
C:\Users\User>ssh sananoorbhatti@192.168.227.129
sananoorbhatti@192.168.227.129's password:
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-31-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Sat Sep 12 08:18:00 PM UTC 2026

System load:  0.0          Processes:           220
Usage of /:   33.8% of 9.75GB Users logged in:          0
Memory usage: 11%         IPv4 address for ens33: 192.168.227.129
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

6 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Last login: Sat Sep 12 20:14:39 2026 from 192.168.227.1
sananoorbhatti@ubuntu:~$ |
```

Figure 41 — windows_ssh_login.png

The SSH connection is successfully established and the Ubuntu Server welcome information is displayed in the Windows terminal.

```
6 updates can be applied immediately.  
To see these additional updates run: apt list --upgradable  
  
Enable ESM Apps to receive additional future security updates.  
See https://ubuntu.com/esm or run: sudo pro status  
  
Last login: Sat Sep 12 20:14:39 2026 from 192.168.227.1  
sananoorbhatti@ubuntu:~$ |
```

Figure 42 — windows_ssh_identity.png

The remote terminal prompt confirms the successful SSH session and displays the required <GitHub-username>@ubuntu identity (sananoorbhatti@ubuntu:~\$).

~ Lab Executed